

Exp: Construct a 2 line to 4 line decoder using basic gates. Construct the active high and active low output decoders.

Objective: To study and construct a 2 line to 4 line decoder using basic logical gates, both with active high output and active low output.

Theory: A decoder is a combinational circuit with n -bit input and m number of mutually exclusive outputs.

The mutually exclusive output means that the decoder circuit makes only one output active (active low or active high) among the m possible outputs and this output is unique, and depends on the input code applied to the n -bit input.

We also know that total input codes available is 2^n . Let the number of outputs be m , then $2^n = m$.

Table 1(a), shows the output and input combinations for 2 line to 4 line decoder with active high outputs, with block diagram.

Table 1(b), shows the output and input combination for 2 line to 4 line decoder with active low outputs, with block diagram

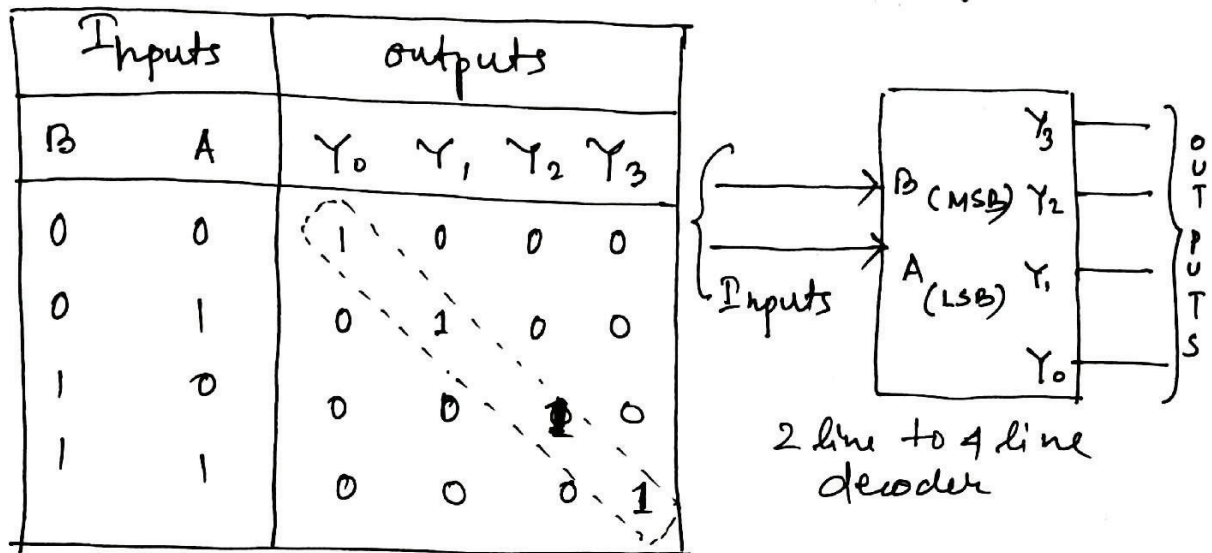


Figure 1(a). Truth table of 2 line 4 line decoder with active high output, with block diagram

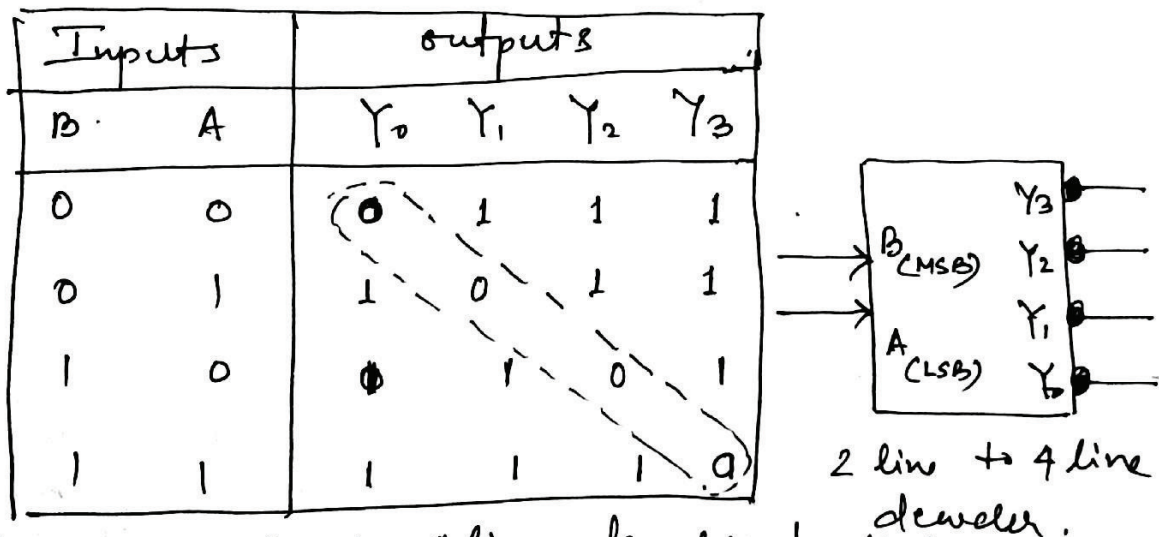


Figure 1(b), 2 line to 4 line decoder truth table with active low outputs and block diagram.

The decoders are also known as minterm generator.

From the above truth table we can find out the minterms generated by the decoders.

Design

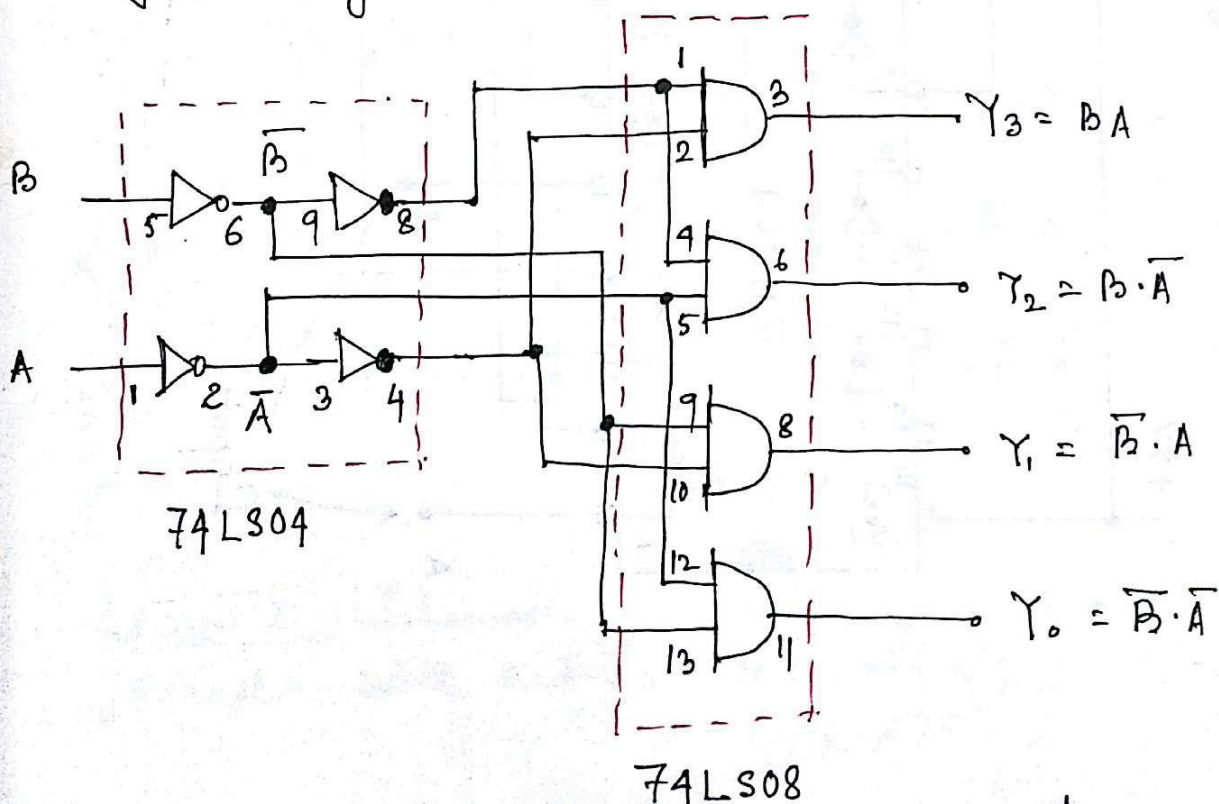
(a) 2 to 4 decoder with active high outputs.

The function table of 2 to 4 decoder is shown in the figure/table 1(a).

Using this function table we can get the min terms as shown below;

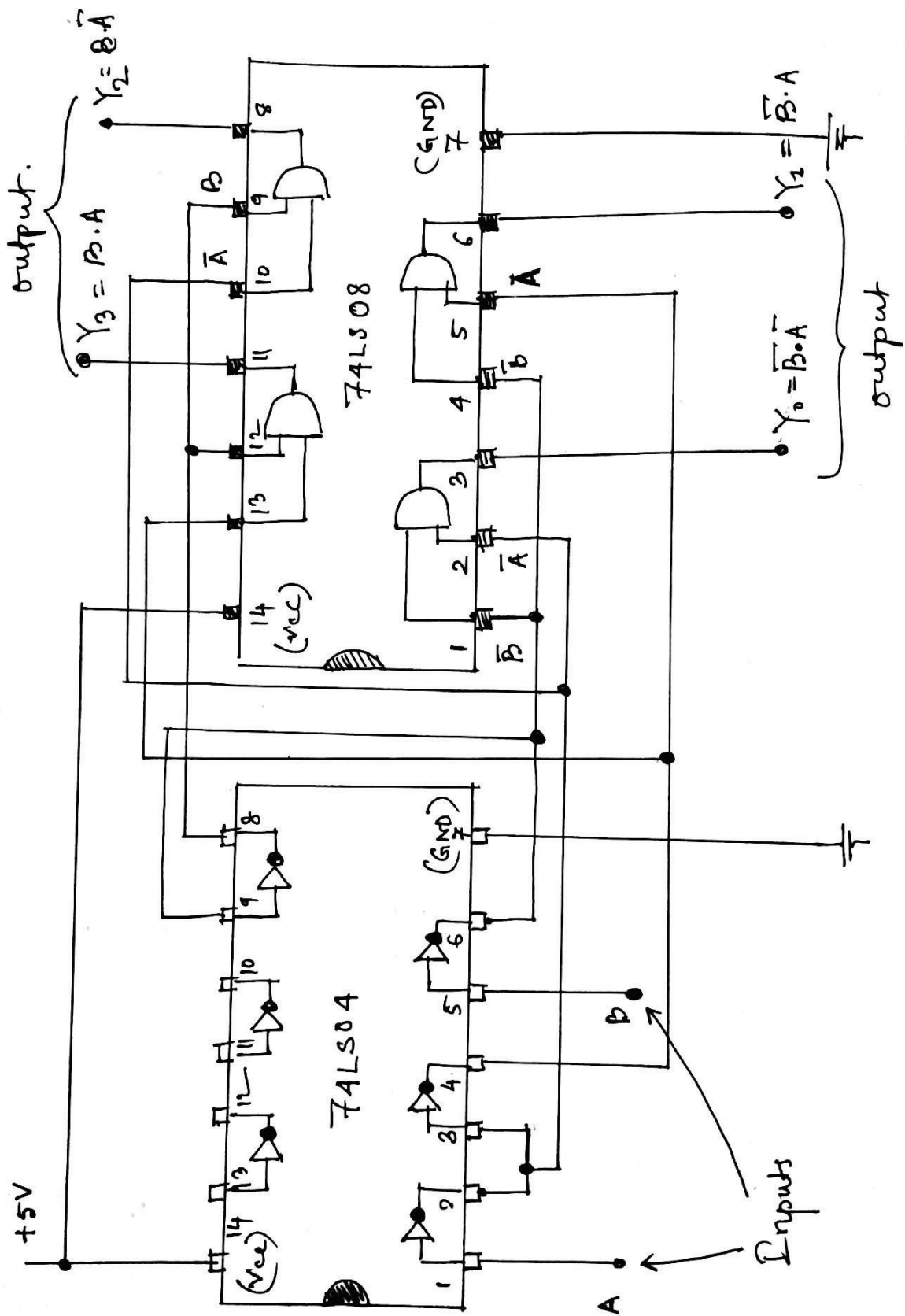
$$\left. \begin{array}{l} Y_0 = \bar{B} \bar{A} \\ Y_1 = \bar{B} A \\ Y_2 = B \bar{A} \\ Y_3 = B A \end{array} \right\} \begin{array}{l} \text{outputs / minterms for decoder with} \\ \text{active high outputs.} \end{array}$$

The Circuit (1) shows the schematic diagram of a active high output decoder using basic gates only.



Circuit (1), shows schematic diagram of 2 to 4 decoder. (3)

Practical Circuit layout of 2 line to 4 line decoder.

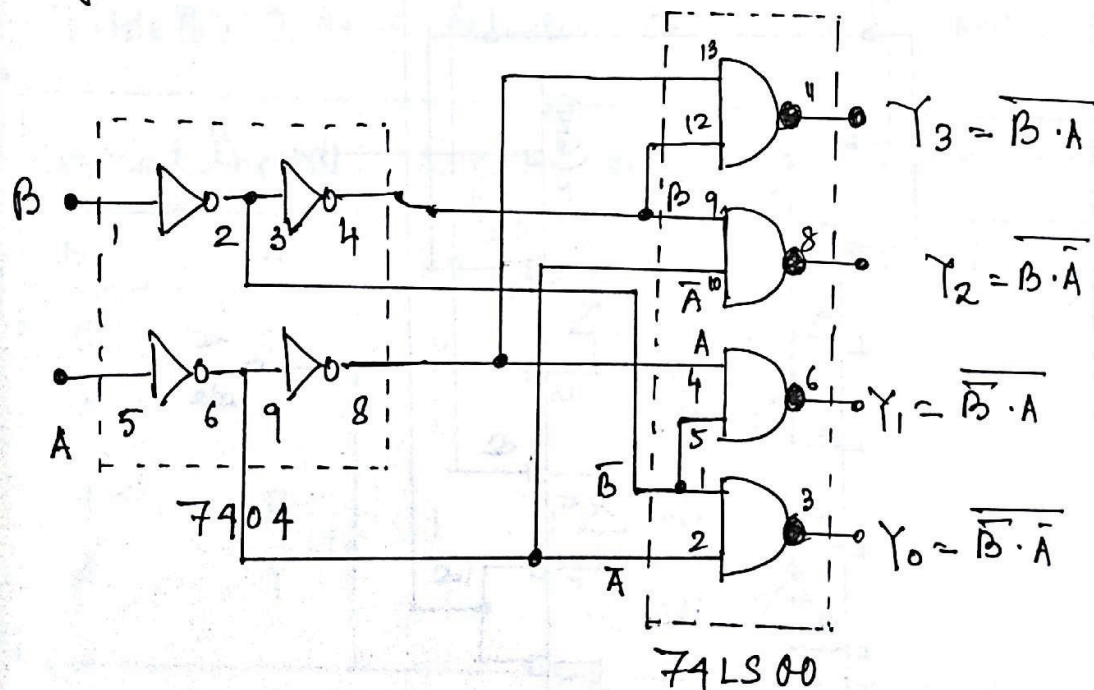


2) 2 to 4 decoder with active low output.

From the functional table 1(b), we can easily write down the minterms for 2 to 4 decoder with active low output.

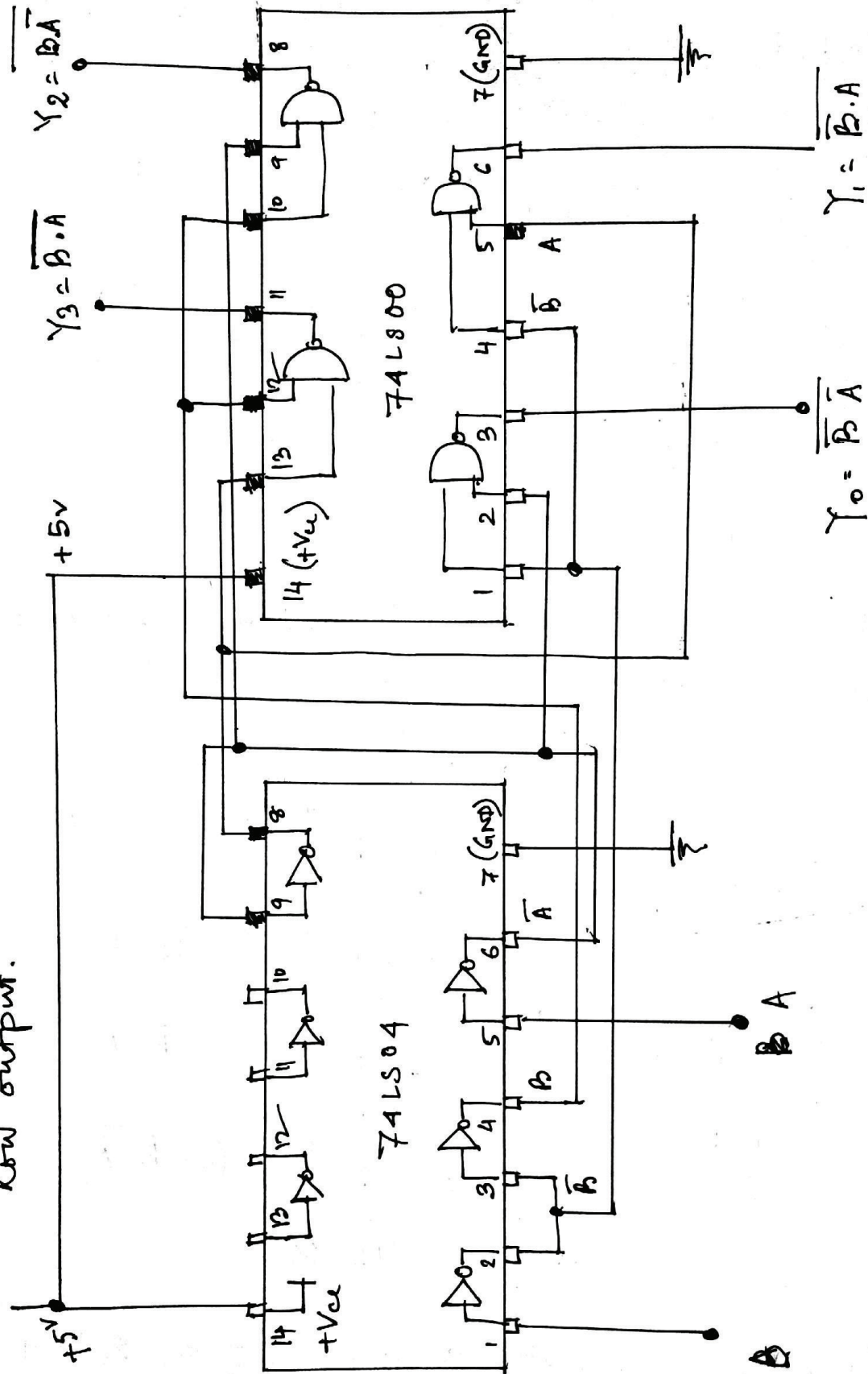
$$\left. \begin{aligned} Y_0 &= \overline{B} \cdot \overline{A} \\ Y_1 &= \overline{B} \cdot A \\ Y_2 &= B \cdot \overline{A} \\ Y_3 &= B \cdot A \end{aligned} \right\} \text{output/minterms of active low outputs.}$$

This can be easily constructed using NAND gates as shown below;



Circuit 2. Schematic diagram of a 2 to 4 decoder with active low outputs.

Practical Circuit Layout of 2 line 4 line decoder with active low output.



Experimental Observation:

(i) Table (a) 2 to 4 decoder with active high output.

Logical Input		Logical output				output voltage			
B	A	Y_0	Y_1	Y_2	Y_3	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0				
0	1	0	1	0	0				
1	0	0	0	1	0				
1	1	0	0	0	1				

ii) Table (b) 2 to 4 decoder with active low output.

Logical Input		Logical output				output voltage			
B	A	Y_0	Y_1	Y_2	Y_3	Y_0	Y_1	Y_2	Y_3
0	0	0	1	1	1				
0	1	1	0	1	1				
1	0	1	1	0	1				
1	1	1	1	1	0				

Precaution/Discussion:

(i) If 74XX Series of TTL Integrated Circuit is used, then a Regulated power Supply of Voltage rating 5V must be used. (7)

- (ii) If LED is used to visualize the output, a current limiting resistor must be used to avoid LED Burnout.
- (iv) Bread board must be checked for internal short circuit.
- (iv) If the chips/ic used are of different TTL family, then fan in/fanout must be checked.
- (v) Floating inputs must be avoided to avoid stray Electromagnetic Interference (noise).
- (vi) Before powering on the circuit, we must thoroughly check for any faulty or short circuit connections.
- (vii) Small jumper/Connecting wires must be used to avoid pick up of stray noise.